

GABRIEL-POPESCU SUPER RINGS

FRANK MURPHY-HERNANDEZ AND FERNANDO CORNEJO-MONTANO

ABSTRACT. The main result of this paper is that the category of supermodules over an arbitrary superring is equivalent to the category of modules over an ordinary ring. The proof is obtained by viewing the category of supermodules as a Grothendieck category and applying the Gabriel–Popescu theorem to a canonical generator. We determine the corresponding endomorphism ring and show that the associated Gabriel topology is trivial, yielding an equivalence of categories rather than merely a localization. This characterization allows the theory of supermodules to be studied entirely within the framework of ordinary module categories and provides a direct connection with Morita theory. We conclude by relating the resulting ring to Clifford algebras and trivial extensions.

INTRODUCTION

Superalgebras are fundamental algebraic structures that extend classical algebraic concepts through the introduction of a $\mathbb{Z}_2$ grading. This grading allows the distinction between "bosonic" and "fermionic" components, resulting in unique algebraic properties and significant applications in various areas of mathematics and theoretical physics. The concept of superalgebras emerged within the framework of quantum field theory and particle physics during the 1970s, when theoretical physicists began exploring symmetries involving both bosonic and fermionic particles. However, the mathematical formalization of superalgebras developed concurrently, largely inspired by the need to extend classical algebraic structures to accommodate these new symmetries. One of the pioneering works in the formalization of superalgebras was by V. G. Kac in 1977 [8], where he introduced Lie superalgebras and established many of the fundamental properties and classifications essential to the theory of superalgebras today. Although Kac's work focused on Lie superalgebras, it laid the groundwork for the broader study of superalgebras.

The theory of superalgebras naturally extends to the field of supergeometry, which is the study of supermanifolds and sheaves over them. In supergeometry, superalgebras are used to define superspaces and supermanifolds, extending classical notions of differential and algebraic geometry to include commutative and anticommutative elements. Supermanifolds, for example, are spaces that locally resemble a graded vector space and are modeled using superalgebras. This extension allows the study of geometric objects that possess both bosonic and fermionic degrees of freedom, providing a unified framework for various physical and mathematical theories. See [4], [11] and [18].

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Instead of the category of vector spaces, where super algebras are typically defined, attention shifts to the category of abelian groups, giving rise to the concept of super rings as their analogous structures. Super rings extend the algebraic framework of super algebras into commutative rings, incorporating a $\mathbb{Z}_2$ -grading that distinguishes elements by parity. This grading enriches the algebraic landscape, facilitating a deeper exploration of symmetry and structure, particularly within the domain of K-theory.

Super rings are pivotal in K-theory, particularly in its topological and complex versions. The theorem of periodicity by Borel-Bott-Weil, foundational in algebraic topology, illustrates how periodicity in K-theory corresponds to properties of super rings. This theorem not only offers profound insights into algebraic and topological relationships but also underscores the fundamental role of super rings in contemporary mathematical research.

Grothendieck categories, named after Alexander Grothendieck, are abelian categories that satisfy specific axioms making them particularly well-behaved and versatile for many mathematical contexts. These categories generalize the properties of module categories and provide a natural setting for a vast range of algebraic and geometric theories. Key properties of Grothendieck categories include the existence of enough injectives, direct limits, and generators, along with satisfying Grothendieck's axioms AB3, AB4, and AB5. Grothendieck categories were introduced by Alexander Grothendieck in his seminal paper [6].

The development of the Gabriel-Popescu Theorem was a natural progression from Grothendieck's pioneering work in the 1950s, which laid the foundation for the theory of abelian categories. Gabriel's work on localization and abelian categories, combined with Popescu's contributions to category theory, culminated in this theorem that elegantly tied Grothendieck categories to module categories in [14].

The Gabriel-Popescu Theorem has had a significant impact on various mathematical domains. In algebraic geometry, Grothendieck categories provide a natural framework for studying coherent sheaves and other geometric structures. The theorem allows mathematicians to leverage the rich theory of module categories to explore these geometric objects. Similarly, in homological algebra, the theorem facilitates the use of module-theoretic techniques to study derived categories and triangulated categories, which are central to modern algebraic geometry and representation theory. Furthermore, the Gabriel-Popescu Theorem has inspired further research and generalizations, extending its applicability to other contexts, such as quasi-coherent sheaves on schemes and categories of sheaves in topos theory. Its influence is evident in the continued exploration and refinement of categorical equivalences in various mathematical settings.

1. PRELIMINARIES

Let R be a ring. We denote by $\mathcal{M}_R$ the category of right R -modules and R -homomorphisms. Since we will focus on one side, we will simply call right modules modules.

A super abelian group is an abelian group A with a decomposition $A = A_0 \oplus A_1$. There is a function $d: A_0 \cup A_1 \rightarrow \mathbb{Z}_2$ given by

$$d(x) = \begin{cases} 0 & \text{if } x \in A_0, \\ 1 & \text{if } x \in A_1. \end{cases}$$

The elements of $A_0 \cup A_1$ are called homogeneous elements, and $d(x)$ is called the degree of x .

Let $A = A_0 \oplus A_1$ and $B = B_0 \oplus B_1$ be super abelian groups, and let $f: A \rightarrow B$ be a group homomorphism. We say that f is a morphism of super abelian groups if $f(A_i) \subseteq B_i$ for $i = 0, 1$. We denote the set of all morphisms of super abelian groups by $\text{sHom}(A, B)$.

Let $s\mathcal{A}$ denote the category of super abelian groups and their morphisms. This category is isomorphic to $\text{Fun}(\mathbf{2}, \mathcal{A})$, where $\mathbf{2} = \{0, 1\}$ is the discrete category with two objects and $\mathcal{A}$ is the category of abelian groups. Moreover, $s\mathcal{A}$ is a monoidal category with the tensor product over $\mathbb{Z}$: if A and B are super abelian groups, then

$$(A \otimes B)_0 = (A_0 \otimes_{\mathbb{Z}} B_0) \oplus (A_1 \otimes_{\mathbb{Z}} B_1),$$

and

$$(A \otimes B)_1 = (A_0 \otimes_{\mathbb{Z}} B_1) \oplus (A_1 \otimes_{\mathbb{Z}} B_0).$$

A superring is a monoid in the monoidal category $(s\mathcal{A}, \otimes)$. Thus a superring R is a ring whose underlying abelian group is a super abelian group satisfying $R_i R_j \subseteq R_{i+j}$ for $i, j \in \mathbb{Z}_2$, and with $1 \in R_0$.

A super right R -module is a super abelian group M together with a right unitary action of R such that

$$M_i R_j \subseteq M_{i+j}$$

for $i, j \in \mathbb{Z}_2$. Let $M = M_0 \oplus M_1$ and $N = N_0 \oplus N_1$ be super right R -modules. A morphism $f: M \rightarrow N$ of super abelian groups is a super R -homomorphism if it is an R -homomorphism. We denote the set of super R -homomorphisms from M to N by $\text{sHom}_R(M, N)$, and the category of super right R -modules and super R -homomorphisms by $s\mathcal{M}_R$. We will suppose that all super modules are right super modules, so we will omit the prefix “right”. In particular,

$$s\mathcal{A} = s\mathcal{M}_{\mathbb{Z}^\circ}.$$

The foundations of superalgebra are thoroughly treated in the works of Deligne and Morgan [4] and Tuyenman [17].

A Grothendieck category $\mathcal{G}$ is a category that satisfies the following conditions:

- $\mathcal{G}$ is an abelian category.
- $\mathcal{G}$ has arbitrary coproducts.
- $\mathcal{G}$ has a generator.
- Direct limits of short exact sequences in $\mathcal{G}$ are exact in $\mathcal{G}$.

A Gabriel topology $\mathfrak{F}$ on a ring R is a filter of right ideals of R such that:

- For $I \in \mathfrak{F}$ and $a \in R$, we have $(I : a) \in \mathfrak{F}$.
- For $J \in \mathfrak{F}$ and a right ideal I of R such that $(I : a) \in \mathfrak{F}$ for all $a \in J$, it follows that $I \in \mathfrak{F}$.

A right module M is called $\mathfrak{F}$ -closed if the canonical morphism

$$\text{Hom}_R(R, M) \rightarrow \text{Hom}_R(I, M)$$

is an isomorphism for every $I \in \mathfrak{F}$. We denote the full subcategory of $\mathcal{M}_R$ consisting of $\mathfrak{F}$ -closed modules by $\mathcal{M}_{(R, \mathfrak{F})}$. For a comprehensive treatment of Gabriel topologies and their properties, we refer to the classical monograph by Stenström [16].

2. THE CATEGORY OF SUPERRINGS

Having established the monoidal structure of super abelian groups and the definition of a superring as a monoid therein, it is natural to examine the relationship between the category of superrings and the classical category of rings. This connection is mediated by two fundamental functors that allow us to transport properties and constructions from the ordinary setting to the graded one, and vice versa.

Definition 2.1. *For a superring R , we denote by $U(R)$ its underlying ring (obtained by forgetting the $\mathbb{Z}_2$ -grading). This assignment is functorial and gives a forgetful functor*

$$U: s\mathcal{R} \longrightarrow \mathcal{R},$$

from the category of superrings to the category of rings.

This forgetful functor strips away the parity information, thereby allowing us to regard any superring as an ordinary ring. However, it is important to note that not every ring homomorphism between the underlying rings lifts to a morphism in $s\mathcal{R}$; for a map to be a superring homomorphism, it must preserve the grading components. The functor U is therefore faithful but not full. Despite this loss of information, U is crucial for transferring classical notions—such as ideals, quotients, and torsion theories—to the super context, provided one keeps track of the compatibility with the grading.

Conversely, there is a canonical way to embed ordinary rings into the super world.

Definition 2.2. *A ring R may be thought of as a superring R° with trivial decomposition $R^\circ = R \oplus 0$.*

It is a common source of confusion to claim that $R \mapsto R^\circ$ is left adjoint to the forgetful functor U that gives the whole underlying ring. This is not the case, since a morphism of superrings $R^\circ \rightarrow S$ must send the even part R into the even part S_0 , whereas a morphism $R \rightarrow U(S)$ only lands in the whole ring S . The correct adjunction is between $(-)^\circ$ and the even-part functor $(-)_0$, as stated in the following proposition.

Proposition 2.1. *The functor*

$$(-)^\circ: \mathcal{R} \longrightarrow s\mathcal{R}, \quad R \longmapsto R^\circ := R \oplus 0,$$

is fully faithful and is left adjoint to the even-part functor

$$(-)_0: s\mathcal{R} \longrightarrow \mathcal{R}, \quad S = S_0 \oplus S_1 \longmapsto S_0.$$

Equivalently, for every ring R and every superring S , there is a natural bijection

$$\mathrm{Hom}_{s\mathcal{R}}(R^\circ, S) \cong \mathrm{Hom}_{\mathcal{R}}(R, S_0).$$

Proof. We verify the adjunction via the triangular identities. Define the unit $\eta: 1_{\mathcal{R}} \rightarrow (-)_0 \circ (-)^\circ$ by

$$\eta_R: R \rightarrow (R^\circ)_0 = R, \quad r \mapsto r,$$

i.e. $\eta_R = \text{id}_R$. Define the counit $\varepsilon: (-)^\circ \circ (-)_0 \rightarrow 1_{s\mathcal{R}}$ by

$$\varepsilon_S: (S_0)^\circ \longrightarrow S, \quad s_0 \mapsto s_0,$$

which is the canonical inclusion of the even part into S . This is a morphism of superrings because it sends the even part of $(S_0)^\circ$ (which is S_0), and the odd part (which is 0) into S_1 .

Now we check the triangular identities.

First identity: For $S \in s\mathcal{R}$,

$$(-)_0(\varepsilon_S) \circ \eta_{(-)_0(S)} = \text{id}_{S_0}.$$

Indeed, $(-)_0(S) = S_0$, so $\eta_{S_0} = \text{id}_{S_0}$. Also, $(-)_0(\varepsilon_S)$ is the even part of the inclusion $S_0 \hookrightarrow S$, which is precisely id_{S_0} . Hence the composition is the identity on S_0 .

Second identity: For $R \in \mathcal{R}$,

$$\varepsilon_{R^\circ} \circ (-)^\circ(\eta_R) = \text{id}_{R^\circ}.$$

Here $R^\circ = R \oplus 0$, so its even part is R . The map $(-)^\circ(\eta_R)$ is the superring morphism induced by $\eta_R = \text{id}_R$, which is the identity on the even part R and zero on the odd part 0. The map ε_{R° is the inclusion of the even part R into R° . Composing them gives the identity on R° .

Since both triangular identities hold, it follows from the standard adjunction theorem that $(-)^\circ$ is left adjoint to $(-)_0$. The fully faithfulness is immediate from the fact that $(-)^\circ$ is injective on objects and the bijection on Hom-sets restricts to an identification of $\text{Hom}_{\mathcal{R}}(R, R')$ with $\text{Hom}_{s\mathcal{R}}(R^\circ, (R')^\circ)$. $\square$

We recall the classical construction of trivial extensions (or idealizations) and show how they naturally give rise to superrings.

Definition 2.3. Let R be an associative ring with unity and let M be an R - R -bimodule. The **trivial extension** (or **idealization**) of R by M , denoted $R \ltimes M$, is the abelian group $R \oplus M$ equipped with the multiplication

$$(r, m)(s, n) := (rs, rn + ms),$$

for all $r, s \in R$ and $m, n \in M$.

It is straightforward to verify that $R \ltimes M$ is an associative ring with unit $(1_R, 0)$. This construction is classical and a comprehensive reference is the survey article *Idealization of a module* by Anderson and Winders [1], where the fundamental properties of this construction are systematically studied. Further references include the books by Nagata [12], Anderson and Fuller [2], and Lam [9].

The trivial extension carries a canonical superring structure.

Proposition 2.2. Let M be an R - R -bimodule. The ring $R \ltimes M$ is a superring with $\mathbb{Z}_2$ -grading given by

$$(R \ltimes M)_0 := R \oplus 0, \quad (R \ltimes M)_1 := 0 \oplus M.$$

Proof. The unit $(1, 0)$ clearly lies in the even part. It remains to check the compatibility of the multiplication with the grading. For $r, s \in R$ and $m, n \in M$, we have

$$\begin{aligned} (r, 0)(s, 0) &= (rs, 0) \in (R \ltimes M)_0, \\ (r, 0)(0, m) &= (0, rm) \in (R \ltimes M)_1, \\ (0, m)(r, 0) &= (0, mr) \in (R \ltimes M)_1, \end{aligned}$$

and finally,

$$(0, m)(0, n) = (0, 0 \cdot n + m \cdot 0) = (0, 0) \in (R \ltimes M)_0.$$

Hence $(R \ltimes M)_i(R \ltimes M)_j \subseteq (R \ltimes M)_{i+j}$ for all $i, j \in \mathbb{Z}_2$, so $R \ltimes M$ is indeed a superring. $\square$

We now show that this construction is natural in the module argument.

Proposition 2.3. *For a fixed ring R , the assignment $M \mapsto R \ltimes M$ defines a functor*

$$T_R: R\text{-}\mathbf{Bimod} \longrightarrow s\mathcal{R},$$

from the category of R - R -bimodules to the category of superrings.

Proof. For a morphism $f: M \rightarrow N$ in $R\text{-}\mathbf{Bimod}$ (i.e., a bimodule homomorphism), define

$$T_R(f): R \ltimes M \longrightarrow R \ltimes N, \quad (r, m) \longmapsto (r, f(m)).$$

We verify that $T_R(f)$ is a morphism of superrings. It is clearly additive and preserves the unit. Moreover, it preserves the multiplication since

$$T_R(f)((r, m)(s, n)) = T_R(f)(rs, rn + ms) = (rs, f(rn + ms)) = (rs, rf(n) + f(m)s),$$

while

$$T_R(f)(r, m)T_R(f)(s, n) = (r, f(m))(s, f(n)) = (rs, rf(n) + f(m)s).$$

The two expressions coincide because f is a bimodule homomorphism. Furthermore,

$$T_R(f)(R \oplus 0) \subseteq R \oplus 0, \quad T_R(f)(0 \oplus M) \subseteq 0 \oplus N,$$

so $T_R(f)$ preserves the grading. Hence $T_R(f)$ is a well-defined morphism in $s\mathcal{R}$.

The identity is preserved since $T_R(\text{id}_M) = \text{id}_{R \ltimes M}$, and for composable morphisms $f: M \rightarrow N$ and $g: N \rightarrow P$, we have

$$T_R(g \circ f)(r, m) = (r, (g \circ f)(m)) = (r, g(f(m))) = T_R(g)(r, f(m)) = (T_R(g) \circ T_R(f))(r, m).$$

Thus T_R is a covariant functor. $\square$

Consequently, every R - R -bimodule gives rise to a superring whose underlying ring is the trivial extension, and this construction is natural with respect to bimodule homomorphisms.

Let R be a fixed ring. We denote by $s\mathcal{R}^{/R}$ the category of superrings over R , i.e., pairs (S, i) where S is a superring and $i: R \rightarrow S_0$ is a morphism of rings (identifying R with a subring of the even part). Morphisms in $s\mathcal{R}^{/R}$ are superring homomorphisms that commute with the structural morphisms from R .

Let $s\mathcal{R}_{\text{sq}}^{/R}$ be the full subcategory of $s\mathcal{R}^{/R}$ consisting of those superrings S whose odd part satisfies

$$S_1^2 = 0,$$

i.e., the product of any two odd elements is zero. In such a superring, the even part S_0 is a ring containing R , and the odd part S_1 is an R - R -bimodule via the actions induced by i .

Define the functor

$$V: s\mathcal{R}_{\text{sq}}^{/R} \longrightarrow R\text{-}\mathbf{Bimod}, \quad V(S) := S_1,$$

where the left and right R -actions on S_1 are given by

$$r \cdot s_1 := i(r)s_1, \quad s_1 \cdot r := s_1 i(r),$$

for $r \in R$ and $s_1 \in S_1$.

On the other hand, we have the trivial extension functor

$$T_R: R\text{-}\mathbf{Bimod} \longrightarrow s\mathcal{R}_{\text{sq}}^{/R}, \quad M \longmapsto R \ltimes M,$$

where $R \ltimes M$ is endowed with the superring structure described in the previous section, with structural morphism $i_R: R \rightarrow (R \ltimes M)_0 = R \oplus 0$ given by $i_R(r) = (r, 0)$.

Proposition 2.4. *The functor T_R is left adjoint to V . That is, for every R - R -bimodule M and every superring $S \in s\mathcal{R}_{\text{sq}}^{/R}$, there is a natural bijection*

$$\text{Hom}_{s\mathcal{R}^{/R}}(R \ltimes M, S) \cong \text{Hom}_{R\text{-}\mathbf{Bimod}}(M, S_1).$$

Proof. We construct the bijection explicitly. Let $i: R \rightarrow S_0$ denote the structural morphism of S .

Given a morphism $f: R \ltimes M \rightarrow S$ in $s\mathcal{R}^{/R}$, we define

$$\Phi(f): M \rightarrow S_1, \quad \Phi(f)(m) := f(0, m).$$

Since f preserves the grading and fixes R , this map lands in S_1 . Moreover, f is a bimodule homomorphism because

$$\Phi(f)(rm) = f(0, rm) = f((r, 0)(0, m)) = f(r, 0)f(0, m) = i(r)\Phi(f)(m),$$

and similarly for the right action.

Conversely, given a bimodule homomorphism $g: M \rightarrow S_1$, define

$$\Psi(g): R \ltimes M \longrightarrow S, \quad \Psi(g)(r, m) := i(r) + g(m).$$

We verify that $\Psi(g)$ is a superring homomorphism. It clearly preserves addition and the unit. For the multiplication:

$$\Psi(g)((r, m)(s, n)) = \Psi(g)(rs, rn + ms) = i(rs) + g(rn) + g(ms).$$

Since i is a ring homomorphism and g is a bimodule homomorphism, this equals

$$i(r)i(s) + i(r)g(n) + g(m)i(s).$$

On the other hand,

$$\Psi(g)(r, m)\Psi(g)(s, n) = (i(r) + g(m))(i(s) + g(n)).$$

Expanding this product in S , we get

$$i(r)i(s) + i(r)g(n) + g(m)i(s) + g(m)g(n).$$

Because $S \in s\mathcal{R}_{\text{sq}}^{/R}$, we have $S_1^2 = 0$, so $g(m)g(n) = 0$. Hence both expressions coincide. Thus $\Psi(g)$ is a morphism of superrings. It is immediate that Ψ is inverse to Φ , and the bijection is natural in both variables. Therefore, $T_R \dashv V$. $\square$

This adjunction can also be verified via the triangular identities. The unit $\eta_M: M \rightarrow V(T_R(M)) = M$ is the identity, and the counit $\varepsilon_S: T_R(V(S)) = R \ltimes S_1 \rightarrow S$ is given by $(r, s_1) \mapsto i(r) + s_1$, which is well-defined precisely because $S_1^2 = 0$. These satisfy the usual identities, confirming the adjunction. For a classical reference on square-zero extensions and their universal property, see [19] or [9].

3. THE CATEGORY OF SUPER ABELIAN GROUPS

Having established the monoidal structure of super abelian groups and their role as the building blocks for superrings, it is now essential to understand their global categorical properties. In particular, since we aim to apply torsion-theoretic methods in the super setting, we need to know that the category of super abelian groups is a Grothendieck category, so that the machinery of Gabriel topologies and localizations is available.

In [6], Grothendieck proved that if $\mathcal{C}$ is a small category and $\mathcal{G}$ is a Grothendieck category, then the functor category $\text{Fun}(\mathcal{C}, \mathcal{G})$ is again a Grothendieck category. Since the category of super abelian groups $s\mathcal{A}$ is equivalent to $\text{Fun}(\mathbf{2}, \mathcal{A})$, where $\mathbf{2}$ is the discrete category with two objects and $\mathcal{A}$ is the category of abelian groups, it follows immediately that $s\mathcal{A}$ is a Grothendieck category.

Moreover, $s\mathcal{A}$ has a very concrete description as a category of modules over a simple ring, which will be crucial for transferring classical results from module theory.

Proposition 3.1. *The category of super abelian groups $s\mathcal{A}$ is isomorphic to the category of modules over the ring $\mathbb{Z} \times \mathbb{Z}$. That is,*

$$s\mathcal{A} \cong \mathcal{M}_{\mathbb{Z} \times \mathbb{Z}}.$$

Proof. The generator of $s\mathcal{A}$ is the super abelian group $\mathbb{Z} \oplus \mathbb{Z}$, where the first copy is placed in degree 0 and the second in degree 1. Its endomorphism ring as a super abelian group is

$$\text{sHom}_{\mathbb{Z}}(\mathbb{Z} \oplus \mathbb{Z}, \mathbb{Z} \oplus \mathbb{Z}) \cong \mathbb{Z} \times \mathbb{Z},$$

since a degree-preserving morphism is determined independently on each homogeneous component.

We now establish the isomorphism explicitly.

Let $A = A_0 \oplus A_1$ be a super abelian group. Define an action of $\mathbb{Z} \times \mathbb{Z}$ on A by

$$(a_0, a_1) \cdot (x_0 + x_1) := a_0 x_0 + a_1 x_1,$$

for $(a_0, a_1) \in \mathbb{Z} \times \mathbb{Z}$, $x_i \in A_i$. This makes A a $\mathbb{Z} \times \mathbb{Z}$ -module.

If $f: A \rightarrow B$ is a morphism of super abelian groups, then for $x = x_0 + x_1 \in A$ we have

$$f((a_0, a_1) \cdot x) = f(a_0 x_0 + a_1 x_1) = a_0 f(x_0) + a_1 f(x_1) = (a_0, a_1) \cdot f(x).$$

Thus f is a $\mathbb{Z} \times \mathbb{Z}$ -module homomorphism.

Conversely, let M be a $\mathbb{Z} \times \mathbb{Z}$ -module. Define

$$M_0 := (1, 0)M, \quad M_1 := (0, 1)M.$$

Since $(1, 0) + (0, 1) = (1, 1)$ is the identity of the ring $\mathbb{Z} \times \mathbb{Z}$, every $m \in M$ decomposes uniquely as

$$m = (1, 1)m = (1, 0)m + (0, 1)m,$$

and the intersection $M_0 \cap M_1$ is zero because $(1, 0)(0, 1)m = (0, 0)m = 0$. Hence $M = M_0 \oplus M_1$, so M is a super abelian group.

If $f: M \rightarrow N$ is a $\mathbb{Z} \times \mathbb{Z}$ -module homomorphism and $x = (1, 0)m \in M_0$, then

$$f(x) = f((1, 0)m) = (1, 0)f(m) \in N_0,$$

and similarly for elements of M_1 . Thus f preserves the grading and is therefore a morphism of super abelian groups.

These assignments are mutually inverse and functorial, yielding the desired isomorphism of categories. $\square$

This result can also be seen as a direct application of the Gabriel–Popescu theorem [15], which states that every Grothendieck category is a localization of a module category over the endomorphism ring of a generator. In this case, the generator $\mathbb{Z} \oplus \mathbb{Z}$ is projective, and the embedding is actually an equivalence (indeed, an isomorphism) with the whole module category. However, the explicit proof given above is more elementary and fully self-contained.

This concrete description of $s\mathcal{A}$ as a module category will be invaluable in the sequel. In particular, it allows us to transport any Gabriel topology or torsion theory from $\mathcal{M}_{\mathbb{Z} \times \mathbb{Z}}$ directly to the super setting, providing a clean bridge between the ordinary and graded worlds. Moreover, the trivial extension construction introduced in the previous section can now be studied in this module-theoretic framework, as the category of super abelian groups is simply the category of modules over a product of two copies of $\mathbb{Z}$.

However, a word of caution is in order regarding the monoidal structures. Although the underlying category of $s\mathcal{A}$ is isomorphic to $\mathcal{M}_{\mathbb{Z} \times \mathbb{Z}}$, the monoidal structure on $s\mathcal{A}$ is given by the tensor product over $\mathbb{Z}$, denoted $\otimes_{\mathbb{Z}}$, whereas the usual tensor product on $\mathcal{M}_{\mathbb{Z} \times \mathbb{Z}}$ is over the ring $\mathbb{Z} \times \mathbb{Z}$, denoted $\otimes_{\mathbb{Z} \times \mathbb{Z}}$. These two monoidal categories are different: the unit object for $(\mathcal{M}_{\mathbb{Z} \times \mathbb{Z}}, \otimes_{\mathbb{Z}})$ is $\mathbb{Z} \oplus 0$ (viewed as a $\mathbb{Z} \times \mathbb{Z}$ -module via the projection onto the first factor), while the unit for $(\mathcal{M}_{\mathbb{Z} \times \mathbb{Z}}, \otimes_{\mathbb{Z} \times \mathbb{Z}})$ is the ring $\mathbb{Z} \times \mathbb{Z}$ itself. This distinction is crucial when studying monoids (i.e., superrings) in these categories, as we did in the previous section.

Despite this difference, the categorical equivalence has a pleasant consequence: the Hom-sets between super abelian groups inherit a natural super abelian group structure. This is expected from the general fact that for a commutative ring R , the category $\mathcal{M}_R$ is an R -category, so its Hom-sets are R -modules. In our case, we obtain the following explicit description.

Proposition 3.2. *Let A and B be super abelian groups. Then $s\text{Hom}(A, B)$ is a super abelian group.*

Proof. Let $A = A_0 \oplus A_1$ and $B = B_0 \oplus B_1$. For any morphism $f \in s\text{Hom}(A, B)$, the condition $f(A_i) \subseteq B_i$ implies that f decomposes uniquely as

$$f = f_0 \oplus f_1,$$

where $f_i: A_i \rightarrow B_i$ is a group homomorphism for $i = 0, 1$. Thus, as abelian groups,

$$s\text{Hom}(A, B) \cong \text{Hom}_{\mathbb{Z}}(A_0, B_0) \oplus \text{Hom}_{\mathbb{Z}}(A_1, B_1).$$

We define the even and odd parts of $s\text{Hom}(A, B)$ by

$$s\text{Hom}(A, B)_0 := \{f \in s\text{Hom}(A, B) \mid f_1 = 0\},$$

and

$$s\text{Hom}(A, B)_1 := \{f \in s\text{Hom}(A, B) \mid f_0 = 0\}.$$

Equivalently, $s\text{Hom}(A, B)_0 \cong \text{Hom}_{\mathbb{Z}}(A_0, B_0)$ and $s\text{Hom}(A, B)_1 \cong \text{Hom}_{\mathbb{Z}}(A_1, B_1)$.

Given any $f = f_0 \oplus f_1$, we can write

$$f = (f_0 \oplus 0) + (0 \oplus f_1),$$

with $f_0 \oplus 0 \in s\text{Hom}(A, B)_0$ and $0 \oplus f_1 \in s\text{Hom}(A, B)_1$. Hence

$$s\text{Hom}(A, B) = s\text{Hom}(A, B)_0 + s\text{Hom}(A, B)_1.$$

Moreover, if $f \in \text{sHom}(A, B)_0 \cap \text{sHom}(A, B)_1$, then $f_0 = 0$ and $f_1 = 0$, so $f = 0$. Thus the intersection is trivial, and we obtain the direct sum decomposition

$$\text{sHom}(A, B) = \text{sHom}(A, B)_0 \oplus \text{sHom}(A, B)_1.$$

Therefore, $\text{sHom}(A, B)$ is indeed a super abelian group. $\square$

Definition 3.1. *Let A and B be super abelian groups. We define their graded tensor product $A \widehat{\otimes} B$ as the super abelian group with components*

$$\begin{aligned} (A \widehat{\otimes} B)_0 &= (A_0 \otimes_{\mathbb{Z}} B_0) \oplus (A_1 \otimes_{\mathbb{Z}} B_1), \\ (A \widehat{\otimes} B)_1 &= (A_0 \otimes_{\mathbb{Z}} B_1) \oplus (A_1 \otimes_{\mathbb{Z}} B_0). \end{aligned}$$

This is exactly the tensor product in the monoidal category $(s\mathcal{A}, \otimes_{\mathbb{Z}})$ defined in the preliminaries.

Proposition 3.3. *For any super abelian group A , the set of super endomorphisms $\text{sHom}(A, A)$ is a superring, with multiplication given by composition of morphisms and with the grading induced by the even/odd decomposition of morphisms defined earlier.*

Proof. We already know from the previous proposition that $\text{sHom}(A, A)$ is a super abelian group. It remains to verify that composition endows it with an associative multiplication that respects the grading, and that the identity morphism is a unit.

Let $f, g \in \text{sHom}(A, A)$. Write $f = f_0 \oplus f_1$ and $g = g_0 \oplus g_1$, where $f_i, g_i: A_i \rightarrow A_i$. The composition $f \circ g$ is given by

$$(f \circ g)_i = f_i \circ g_i: A_i \rightarrow A_i.$$

Thus, if both f and g are even (i.e., their odd components are zero), then $f \circ g$ is even; if one is even and the other odd, the composition is odd; and if both are odd, the composition is even (since $f_1 \circ g_1$ maps A_1 to A_1 , which is even). In all cases, the degree of the composition is the sum of the degrees modulo 2. Hence composition respects the $\mathbb{Z}_2$ -grading.

The composition is associative because it is the usual composition of functions, and it is distributive over addition. The identity morphism id_A is even (since it preserves each homogeneous component) and acts as a two-sided unit for composition. Therefore, $\text{sHom}(A, A)$ is a superring. $\square$

The proposition above shows that $\text{sHom}(A, A)$ is always a superring. The earlier stated condition "if and only if $A = 0$ " would be true only if we required the superring to have, for example, no nonzero odd elements and no nontrivial zero divisors, or if we imposed some other extra condition. In the standard theory, however, the endomorphism superring is a basic and always available example of a superring.

4. THE CATEGORY OF SUPERMODULES

Having established that the category of super abelian groups is a Grothendieck category and, moreover, is isomorphic to the module category over $\mathbb{Z} \times \mathbb{Z}$, we now turn our attention to modules over an arbitrary superring R . This is the natural setting for the torsion-theoretic methods we aim to develop.

Recall that in the ordinary module category $\mathcal{M}_R$, the ring R itself is always a generator: every module is a quotient of a direct sum of copies of R . In the super setting, however, the situation is slightly more subtle. The issue is that a super

R -morphism from R to a supermodule M is determined by the image of the unit $1 \in R_0$, and this image must lie in the even part M_0 . Consequently, the Hom-set $\text{sHom}_R(R, M)$ only captures the even part of M , not the odd part.

Lemma 4.1. *For any $M \in s\mathcal{M}_R$, we have a natural isomorphism*

$$\text{sHom}_R(R, M) \cong M_0,$$

where the isomorphism sends a morphism $f: R \rightarrow M$ to $f(1)$.

Proof. A super R -morphism $f: R \rightarrow M$ is determined by $f(1)$, since for any $r \in R$ we have $f(r) = rf(1)$. Moreover, since f preserves the grading and $1 \in R_0$, we must have $f(1) \in M_0$. Conversely, any $m \in M_0$ defines a super R -morphism $f_m(r) := rm$. Thus the correspondence is bijective and natural in M . $\square$

This observation shows that R alone is not sufficient to generate the entire category $s\mathcal{M}_R$. To obtain a generator, we need to also account for the odd part. The following construction provides the missing piece.

Definition 4.1. *Let $M \in s\mathcal{M}_R$. We define the **parity-twisted supermodule** M^* (also denoted M^σ in the literature) by*

$$M_i^* := M_{1-i}$$

for $i \in \mathbb{Z}_2$. That is, the grading is reversed: the even part of M^* is the odd part of M , and vice versa.

The module structure on M^* is defined using the superring multiplication. Since $R_i M_j^* = R_i M_{1-j} \subseteq M_{i+1-j} = M_{i+j}^*$, we obtain the following.

Proposition 4.1. *For any $M \in s\mathcal{M}_R$, the parity-twisted module M^* is again a right super R -module.*

Proof. Let $r_i \in R_i$ and $x_j \in M_j^* = M_{1-j}$ with $i, j \in \mathbb{Z}_2$. Then

$$r_i x_j \in R_i M_{1-j} \subseteq M_{i+(1-j)} = M_{1-(i+j)} = M_{i+j}^*.$$

Thus the action is compatible with the grading, and the remaining module axioms follow immediately from those of M . $\square$

We now have access to both the even and odd parts of any supermodule via Hom-sets.

Corollary 4.1. *For any $M \in s\mathcal{M}_R$,*

$$\text{sHom}_R(R^*, M) \cong M_1.$$

Proof. By the definition of R^* , we have $R_0^* = R_1$ and $R_1^* = R_0$. A morphism $f: R^* \rightarrow M$ is determined by $f(1)$, but now $1 \in R_1^*$, so $f(1) \in M_1$. Hence the Hom-set identifies with M_1 . $\square$

Corollary 4.2. *For any $M \in s\mathcal{M}_R$,*

$$\text{sHom}_R(R \oplus R^*, M) \cong M_0 \oplus M_1 \cong M,$$

as abelian groups.

Proof. Since Hom preserves direct sums in the first variable, we have

$$\text{sHom}_R(R \oplus R^*, M) \cong \text{sHom}_R(R, M) \oplus \text{sHom}_R(R^*, M) \cong M_0 \oplus M_1 \cong M.$$

The isomorphism is natural in M . $\square$

Corollary 4.3. *The supermodule $R \oplus R^*$ is a generator of the category $s\mathcal{M}_R$.*

Proof. The preceding corollary shows that for every $M \in s\mathcal{M}_R$, there is an epimorphism

$$\bigoplus_{m \in M} (R \oplus R^*) \longrightarrow M,$$

where the copy corresponding to $m \in M$ maps the generator to m . Since this holds for every M , the object $R \oplus R^*$ is a generator. $\square$

With a generator in hand, we can now establish the categorical structure of $s\mathcal{M}_R$.

Proposition 4.2. *Let R be a superring. Then the category $s\mathcal{M}_R$ of right super R -modules is a Grothendieck category.*

Proof. We verify the four axioms of a Grothendieck category.

First, $s\mathcal{M}_R$ is an abelian category. The zero object is the trivial module 0, kernels and cokernels exist and are computed componentwise (as in the ordinary module case), and every monomorphism is a kernel of its cokernel and every epimorphism is a cokernel of its kernel. This follows from the fact that the underlying functor $s\mathcal{M}_R \rightarrow \mathcal{M}_{U(R)}$ (where $U(R)$ is the underlying ordinary ring) is faithful, exact, and reflects isomorphisms.

Second, $s\mathcal{M}_R$ has arbitrary coproducts: they are given by componentwise direct sums of the underlying super abelian groups, with the action of R defined componentwise. These coproducts are also biproducts, as in any abelian category.

Third, as shown in the corollary above, $R \oplus R^*$ is a generator.

Fourth, direct limits (filtered colimits) of short exact sequences in $s\mathcal{M}_R$ are exact. This follows from the fact that filtered colimits commute with finite limits in the category of abelian groups, and the action of R is compatible with the colimit construction. More precisely, the forgetful functor $s\mathcal{M}_R \rightarrow \mathcal{M}_{U(R)}$ preserves and reflects limits and filtered colimits, and the latter is known to be a Grothendieck category.

Therefore, by the Grothendieck axioms, $s\mathcal{M}_R$ is a Grothendieck category. For a detailed treatment of these facts in the graded context, we refer to the comprehensive monograph by Năstăsescu and Van Oystaeyen [13]. $\square$

The fact that $s\mathcal{M}_R$ is a Grothendieck category is essential for the development of torsion theories and localizations in the super context. In particular, the Gabriel–Popescu theorem [15] applies, and the generator $R \oplus R^*$ allows us to represent $s\mathcal{M}_R$ as a localization of a module category over the endomorphism ring of this generator. This provides a powerful bridge between the super and ordinary settings, which will be exploited in the following sections.

5. GABRIEL–POPESCU THEOREM

In the preceding section, we established that for any superring R , the category $s\mathcal{M}_R$ of right super R -modules is a Grothendieck category, and that it admits a canonical generator $G := R \oplus R^*$, where R^* denotes the parity-twisted module. This puts us in a position to apply one of the most powerful tools in the theory of Grothendieck categories: the Gabriel–Popescu theorem. This theorem provides a concrete module-theoretic representation of any Grothendieck category, allowing us to transfer results from ordinary module theory to more general settings.

We now recall the precise statement of the theorem and its consequences for our super context.

Theorem 5.1 (Gabriel–Popescu [?, 15]). *Let $\mathcal{G}$ be a Grothendieck category with a generator G . Let $R := \text{End}_{\mathcal{G}}(G)^{\text{op}}$ be the opposite ring of the endomorphism ring of G (so that $\mathcal{M}_R$ is the category of right R -modules). Consider the functor*

$$T := \text{Hom}_{\mathcal{G}}(G, -): \mathcal{G} \longrightarrow \mathcal{M}_R,$$

which sends an object $X \in \mathcal{G}$ to the right R -module $\text{Hom}_{\mathcal{G}}(G, X)$, with the action of R given by precomposition.

Then:

- (1) *T is fully faithful.*
- (2) *The essential image of T consists precisely of the $\mathfrak{F}$ -closed right R -modules, where $\mathfrak{F}$ is the strongest Gabriel topology on R for which every module of the form $T(X)$, with $X \in \mathcal{G}$, is $\mathfrak{F}$ -closed.*

Consequently, T induces an equivalence of categories

$$\mathcal{G} \simeq \mathcal{M}_{(R, \mathfrak{F})},$$

where $\mathcal{M}_{(R, \mathfrak{F})}$ denotes the full subcategory of $\mathfrak{F}$ -closed modules.

The Gabriel topology $\mathfrak{F}$ constructed in the theorem is sometimes called the **Gabriel topology associated to the generator G** . It is a fundamental invariant of the Grothendieck category $\mathcal{G}$ relative to the chosen generator.

We now compute explicitly the endomorphism ring of the canonical generator $G = R \oplus R^*$ of the category $s\mathcal{M}_R$. This computation is essential for the application of the Gabriel–Popescu theorem, as it provides the ordinary ring over which the supermodule category is represented.

Proposition 5.1. *Let R be a superring. Then*

$$\text{sEnd}_R(R \oplus R^*) \cong \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix},$$

where the multiplication on the right-hand side is given by the usual matrix multiplication, with the entries multiplied according to the superring product of R . In particular, the even part of this endomorphism ring is the diagonal subring $R_0 \oplus R_0$.

Proof. Let $\alpha: R \oplus R^* \rightarrow R \oplus R^*$ be a super R -morphism. Since the domain and codomain are direct sums, α is uniquely determined by four super R -morphisms:

$$\begin{aligned} \alpha_{00}: R &\longrightarrow R, \\ \alpha_{01}: R &\longrightarrow R^*, \\ \alpha_{10}: R^* &\longrightarrow R, \\ \alpha_{11}: R^* &\longrightarrow R^*, \end{aligned}$$

so that in matrix notation,

$$\alpha = \begin{pmatrix} \alpha_{00} & \alpha_{01} \\ \alpha_{10} & \alpha_{11} \end{pmatrix}.$$

We now determine each component explicitly.

First, consider $\alpha_{00}: R \rightarrow R$. Since α_{00} is a super R -morphism, it is determined by $\alpha_{00}(1)$, where $1 \in R_0$. Moreover, the condition $\alpha_{00}(R_i) \subseteq R_i$ forces $\alpha_{00}(1) \in R_0$. Thus α_{00} corresponds uniquely to an element $a_{00} := \alpha_{00}(1) \in R_0$, and for any $r \in R$,

$$\alpha_{00}(r) = ra_{00}.$$

Similarly, $\alpha_{11}: R^* \rightarrow R^*$ is determined by $\alpha_{11}(1)$ with $1 \in R_1^*$. Since α_{11} preserves the grading, we must have $\alpha_{11}(1) \in R_1^* = R_0$. Hence there exists a unique $a_{11} \in R_0$ such that for any $x \in R^*$,

$$\alpha_{11}(x) = xa_{11}.$$

Now consider $\alpha_{01}: R \rightarrow R^*$. This morphism sends $1 \in R_0$ to an element of $R_0^* = R_1$. Thus α_{01} is determined by a unique element $a_{01} := \alpha_{01}(1) \in R_1$, and for any $r \in R$,

$$\alpha_{01}(r) = ra_{01}.$$

Finally, consider $\alpha_{10}: R^* \rightarrow R$. Since $1 \in R_1^*$ and α_{10} must send $R_1^* = R_0$ to R_1 , we have $\alpha_{10}(1) \in R_1$. Thus there exists a unique $a_{10} \in R_1$ such that for any $x \in R^*$,

$$\alpha_{10}(x) = xa_{10}.$$

Therefore, every super R -morphism α corresponds uniquely to a matrix

$$\begin{pmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{pmatrix} \quad \text{with} \quad a_{00}, a_{11} \in R_0, \quad a_{01}, a_{10} \in R_1.$$

Conversely, any such matrix defines a super R -morphism by the formulas above. Hence the assignment is bijective.

It remains to verify that composition of endomorphisms corresponds to matrix multiplication. Let

$$\alpha = \begin{pmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{pmatrix}, \quad \beta = \begin{pmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{pmatrix}$$

be two endomorphisms. Then $\beta \circ \alpha$ is represented by the matrix whose (i, j) -entry is determined by the composition of the corresponding components. For example, the $(0, 0)$ -entry is

$$(\beta \circ \alpha)_{00} = \beta_{00} \circ \alpha_{00} + \beta_{01} \circ \alpha_{10}.$$

For $r \in R$,

$$(\beta_{00} \circ \alpha_{00})(r) = \beta_{00}(ra_{00}) = (ra_{00})b_{00} = r(a_{00}b_{00}),$$

and

$$(\beta_{01} \circ \alpha_{10})(r) = \beta_{01}(ra_{10}) = (ra_{10})b_{01} = r(a_{10}b_{01}),$$

since $ra_{10} \in R_1$ and $b_{01} \in R_1$, so the product is well-defined in R . Thus the $(0, 0)$ -entry is $a_{00}b_{00} + a_{10}b_{01}$.

Similarly, the other entries are:

$$(\beta \circ \alpha)_{01} = a_{00}b_{01} + a_{10}b_{11},$$

$$(\beta \circ \alpha)_{10} = a_{01}b_{00} + a_{11}b_{10},$$

$$(\beta \circ \alpha)_{11} = a_{01}b_{01} + a_{11}b_{11}.$$

These are precisely the entries of the product of the matrices

$$\begin{pmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{pmatrix} \begin{pmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{pmatrix} = \begin{pmatrix} a_{00}b_{00} + a_{01}b_{10} & a_{00}b_{01} + a_{01}b_{11} \\ a_{10}b_{00} + a_{11}b_{10} & a_{10}b_{01} + a_{11}b_{11} \end{pmatrix},$$

where the products $a_{ij}b_{kl}$ are taken in the superring R . Hence composition corresponds exactly to matrix multiplication.

The identity morphism corresponds to the identity matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

with $1 \in R_0$ and $0 \in R_1$. Therefore, we have an isomorphism of rings

$$\mathrm{sEnd}_R(R \oplus R^*) \cong \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix}.$$

Finally, the even part of this endomorphism ring consists of those matrices that preserve the parity of the morphism, i.e., those with $a_{01} = a_{10} = 0$. Thus the even part is isomorphic to $R_0 \times R_0$, the diagonal subring. $\square$

This is the ordinary ring over which the category $s\mathcal{M}_R$ is represented as a category of $\mathfrak{F}$ -closed modules. Specifically, the functor

$$T = \mathrm{sHom}_R(R \oplus R^*, -): s\mathcal{M}_R \longrightarrow \mathcal{M}_S$$

is fully faithful, and its essential image consists of the S -modules that are closed with respect to the induced Gabriel topology $\mathfrak{F}$.

If R is a superring with trivial grading (i.e., $R_1 = 0$), then the endomorphism ring reduces to

$$S \cong \begin{pmatrix} R & 0 \\ 0 & R \end{pmatrix} \cong R \times R,$$

and the generator $R \oplus R^*$ is simply $R \oplus R$, which is the usual generator of the category $\mathcal{M}_{R \times R}$. This recovers the classical situation.

We saw that the category of supermodules $s\mathcal{M}_R$ is a Grothendieck category and that $G = R \oplus R^*$ is a generator. We now prove a much stronger statement: G is not just a generator, but a small projective generator. This has profound consequences for the Gabriel–Popescu representation, as it implies that the associated Gabriel topology is trivial and that $s\mathcal{M}_R$ is actually Morita equivalent to a category of ordinary modules.

Proposition 5.2. *Let R be a superring and let $G = R \oplus R^* \in s\mathcal{M}_R$. Then:*

- (1) *G is a projective object in $s\mathcal{M}_R$.*
- (2) *G is small (finitely generated), i.e., the functor $\mathrm{sHom}_R(G, -)$ commutes with arbitrary direct sums.*

Proof. We prove each statement separately.

(1) Projectivity: First, consider R as a right supermodule over itself. Since R is a superring with unity $1 \in R_0$, the module R_R is free with basis $\{1\}$ (concentrated in degree 0). Free modules are projective in any module category, so R is projective in $s\mathcal{M}_R$.

Now, consider the parity shift functor

$$(-)^*: s\mathcal{M}_R \longrightarrow s\mathcal{M}_R, \quad M \longmapsto M^*,$$

where $(M^*)_i = M_{1-i}$. This functor is an automorphism of the category: its inverse is itself, since $(M^*)^* = M$. Automorphisms of categories preserve all categorical properties, including projectivity. Therefore, since R is projective, its image under this automorphism, namely R^* , is also projective.

Finally, the direct sum of two projective objects is projective. Hence

$$G = R \oplus R^*$$

is a projective object in $s\mathcal{M}_R$.

(2) Smallness: We show that G is finitely generated. The module R is generated by the single element $1 \in R_0$ (since every $r \in R$ is $r \cdot 1$). Similarly, R^* is generated by the single element $1 \in R_1^*$ (since R^* has the same underlying elements

as R , but with reversed grading; every $x \in R^*$ is $x \cdot 1$). Therefore, both R and R^* are finitely generated.

A finite direct sum of finitely generated objects is finitely generated. Thus $G = R \oplus R^*$ is finitely generated.

In any Grothendieck category (and in particular in module categories), every finitely generated object is small. This means that for any family $\{M_\lambda\}_{\lambda \in \Lambda}$ of objects in $s\mathcal{M}_R$, the canonical morphism

$$\bigoplus_{\lambda} \text{sHom}_R(G, M_\lambda) \longrightarrow \text{sHom}_R\left(G, \bigoplus_{\lambda} M_\lambda\right)$$

is an isomorphism. Hence G is small. $\square$

We now apply the Gabriel–Popescu theorem. Let

$$S := \text{sEnd}_R(G)^{\text{op}}$$

be the endomorphism ring of the generator G , with multiplication reversed so that $s\mathcal{M}_R$ maps to right S -modules. From Proposition 5.1, we know that

$$S \cong \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix}.$$

Consider the functor

$$T := \text{sHom}_R(G, -) : s\mathcal{M}_R \longrightarrow \mathcal{M}_S,$$

which sends a supermodule M to the right S -module $\text{sHom}_R(G, M)$.

Theorem 5.2. *The functor*

$$T : s\mathcal{M}_R \longrightarrow \mathcal{M}_S$$

is an equivalence of categories. Consequently, the associated Gabriel topology $\mathfrak{F}$ on S is the trivial topology, i.e.,

$$\mathfrak{F} = \{I \subseteq S \mid I \text{ is a right ideal of } S\}.$$

Proof. By the Gabriel–Popescu theorem [15], the functor T is fully faithful and its essential image consists precisely of the $\mathfrak{F}$ -closed S -modules for some Gabriel topology $\mathfrak{F}$. It remains to show that every S -module is $\mathfrak{F}$ -closed.

Since G is projective, the functor $T = \text{sHom}_R(G, -)$ is exact: it sends short exact sequences in $s\mathcal{M}_R$ to short exact sequences in $\mathcal{M}_S$.

Since G is a generator, T is faithful: if $f : M \rightarrow N$ is a morphism such that $T(f) = 0$, then $f \circ h = 0$ for every $h : G \rightarrow M$. Because G generates M , this implies $f = 0$.

Since G is small, T preserves arbitrary coproducts (direct sums).

Now, consider the left adjoint of T , which exists because G is a generator. It is given by

$$L : \mathcal{M}_S \longrightarrow s\mathcal{M}_R, \quad L(N) = N \otimes_S G.$$

The unit of the adjunction $\eta : \text{id}_{s\mathcal{M}_R} \rightarrow L \circ T$ is an isomorphism because G is a generator. The counit $\varepsilon : T \circ L \rightarrow \text{id}_{\mathcal{M}_S}$ is also an isomorphism because G is projective (this is the standard characterization of Morita contexts for projective generators). Therefore, T and L form an adjoint equivalence of categories.

Thus the essential image of T is the entire category $\mathcal{M}_S$. Since this image is exactly the subcategory of $\mathfrak{F}$ -closed modules, we have

$$\mathcal{M}_{(S, \mathfrak{F})} = \mathcal{M}_S.$$

The only Gabriel topology for which all modules are closed is the trivial topology, where every right ideal is in $\mathfrak{F}$. Therefore,

$$\mathfrak{F} = \{I \subseteq S \mid I \text{ is a right ideal of } S\}.$$

□

Corollary 5.1. *There is a Morita equivalence*

$$s\mathcal{M}_R \simeq \mathcal{M}_S,$$

where

$$S = \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix}.$$

In particular, the categories of supermodules over R and ordinary modules over the matrix ring S have equivalent abelian structures, and all homological, torsion-theoretic, and categorical properties are preserved under this equivalence.

The fact that the topology is trivial means that no localization or torsion theory is needed to pass between $s\mathcal{M}_R$ and $\mathcal{M}_S$. The equivalence is global, not just a localizing subcategory. This greatly simplifies the study of supermodules, as we can now transfer any result from ordinary module theory over S directly to the super setting, and vice versa, without the need for $\mathfrak{F}$ -closure conditions.

If R is a superring with trivial odd part ($R_1 = 0$), then $R^* = R$ and

$$S \cong \begin{pmatrix} R & 0 \\ 0 & R \end{pmatrix} \cong R \times R.$$

In this case, the equivalence recovers the classical isomorphism $s\mathcal{M}_R \cong \mathcal{M}_{R \times R}$, which is precisely the decomposition of a supermodule into its even and odd parts. This is consistent with the fact that $s\mathcal{A} \cong \mathcal{M}_{\mathbb{Z} \times \mathbb{Z}}$ established earlier.

6. CLIFFORD ALGEBRAS AND THE PARITY MATRIX RING

In this section, we briefly recall the notion of a Clifford algebra and show how the parity matrix ring $S = \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix}$ naturally arises as a Clifford algebra when the odd part R_1 is equipped with a suitable bilinear form. This provides a deeper algebraic interpretation of the Morita equivalence $s\mathcal{M}_R \simeq \mathcal{M}_S$ established in the previous section.

Definition 6.1 (Clifford Algebra). *Let K be a commutative ring, V a K -module, and $Q : V \rightarrow K$ a quadratic form. The **Clifford algebra** $\text{Cl}(V, Q)$ is the quotient of the tensor algebra $T(V)$ by the two-sided ideal generated by the elements*

$$v \otimes v - Q(v) \cdot 1, \quad v \in V.$$

Equivalently, if $B : V \times V \rightarrow K$ is the symmetric bilinear form associated to Q (given by polarization), then $\text{Cl}(V, Q)$ is the associative K -algebra with unit generated by V subject to the relations

$$vw + wv = 2B(v, w), \quad v, w \in V.$$

The Clifford algebra is $\mathbb{Z}_2$ -graded: the even part consists of products of an even number of elements of V , and the odd part consists of products of an odd number of elements. This grading is fundamental for the connection with superalgebras.

Example 6.1. *The simplest examples of Clifford algebras over $K = \mathbb{R}$ are:*

- If $V = \mathbb{R}$ with $Q(x) = x^2$, then $\text{Cl}(\mathbb{R}, Q) \cong \mathbb{R}[x]/(x^2 - 1) \cong \mathbb{R} \times \mathbb{R}$.
- If $V = \mathbb{R}$ with $Q(x) = -x^2$, then $\text{Cl}(\mathbb{R}, Q) \cong \mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.
- If $V = \mathbb{R}^2$ with $Q(x_1, x_2) = x_1^2 + x_2^2$, then $\text{Cl}(\mathbb{R}^2, Q)$ is the algebra of quaternions $\mathbb{H}$.

These examples show that Clifford algebras generalize complex numbers and quaternions.

Now, suppose that R_0 is a commutative ring and that R_1 is a projective R_0 -module of finite rank equipped with a symmetric R_0 -bilinear form $B : R_1 \times R_1 \rightarrow R_0$. Then we can form the Clifford algebra $\text{Cl}(R_1, B)$ over R_0 . This Clifford algebra has a natural $\mathbb{Z}_2$ -grading, and its even part is isomorphic to R_0 .

When the form B is non-degenerate and the Clifford algebra is central simple over R_0 , it admits a matrix representation of the form

$$\text{Cl}(R_1, B) \cong \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix}.$$

In this representation, the even part corresponds to the diagonal matrices $\begin{pmatrix} R_0 & 0 \\ 0 & R_0 \end{pmatrix} \cong R_0 \times R_0$, and the odd part corresponds to the off-diagonal matrices $\begin{pmatrix} 0 & R_1 \\ R_1 & 0 \end{pmatrix} \cong R_1 \oplus R_1$.

Thus, the ring

$$S = \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix}$$

is precisely the Clifford algebra associated to the pair (R_0, R_1) when R_1 is endowed with a suitable bilinear form. This is why S is called the **parity matrix ring** or the **Clifford algebra** associated to the superring R .

The interpretation of S as a Clifford algebra is most transparent when R_0 is commutative and R_1 is a finitely generated projective module with a non-degenerate symmetric bilinear form. In more general situations—for instance, when R_0 is non-commutative or R_1 does not carry such a form—the ring S is still called the parity matrix ring by analogy, but its Clifford algebra structure may be less immediate.

The results obtained in the preceding sections, particularly the Morita equivalence

$${}_S\mathcal{M}_R \simeq \mathcal{M}_S, \quad S = \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix},$$

find a natural and elegant explanation within the classical theory of Clifford algebras. To see this, we first recall a fundamental fact.

Proposition 6.1. *Every Clifford algebra $\text{Cl}(V, Q)$ is naturally a superalgebra (i.e., a $\mathbb{Z}_2$ -graded algebra).*

Proof. Let $\text{Cl}(V, Q)$ be the Clifford algebra associated to a quadratic space (V, Q) over a commutative ring K . By definition, $\text{Cl}(V, Q)$ is the quotient of the tensor

algebra $T(V)$ by the relations $v^2 = Q(v)$. The tensor algebra $T(V)$ is $\mathbb{Z}$ -graded by tensor length:

$$T(V) = \bigoplus_{n \geq 0} V^{\otimes n}.$$

This $\mathbb{Z}$ -grading induces a $\mathbb{Z}_2$ -grading by reduction modulo 2:

$$T(V)_0 = \bigoplus_{n \text{ even}} V^{\otimes n}, \quad T(V)_1 = \bigoplus_{n \text{ odd}} V^{\otimes n}.$$

The defining relations $v^2 = Q(v)$ are homogeneous of degree 0 (since v^2 has degree $2 \equiv 0 \pmod{2}$ and $Q(v)$ has degree 0). Hence the ideal generated by these relations is homogeneous with respect to this $\mathbb{Z}_2$ -grading. Therefore, the quotient $\text{Cl}(V, Q)$ inherits a $\mathbb{Z}_2$ -grading, making it a superalgebra. Explicitly, the even part consists of products of an even number of vectors from V , and the odd part consists of products of an odd number of vectors. $\square$

This is a standard result in the theory of Clifford algebras. For a detailed exposition, see, for instance, the work of Leites [10], where Clifford algebras are studied precisely as superalgebras, or the comprehensive treatment in [18]. The natural $\mathbb{Z}_2$ -grading is also highlighted in modern references such as [3] and in the context of Clifford superalgebras in [5].

Now, suppose that R_0 is a commutative ring and that R_1 is a projective R_0 -module equipped with a symmetric R_0 -bilinear form $B : R_1 \times R_1 \rightarrow R_0$. Then the associated Clifford algebra $\text{Cl}(R_1, B)$ is a superalgebra whose even part is isomorphic to R_0 . Under suitable non-degeneracy conditions, this Clifford algebra admits a matrix representation of the form

$$\text{Cl}(R_1, B) \cong \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix},$$

which is precisely the ring S appearing in our Morita equivalence.

Thus, the general result we have established,

$$s\mathcal{M}_R \simeq \mathcal{M}_S,$$

can be seen as a far-reaching extension of the classical Morita theory of Clifford algebras. In the classical setting, one considers a superring R whose even part R_0 is commutative and whose odd part R_1 is a module over R_0 with a bilinear form. In that case, the equivalence reduces to the well-known fact that the category of modules over a Clifford algebra is Morita equivalent to the category of modules over its even part (or to a suitable matrix ring). Our result shows that this phenomenon is not limited to the commutative, bilinear-form case: it holds for *any* superring R , without any additional assumptions on R_0 or R_1 . The parity matrix ring S always provides the Morita equivalent ordinary ring.

Corollary 6.1. *The Morita equivalence*

$$s\mathcal{M}_R \simeq \mathcal{M} \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix}$$

extends the classical Morita equivalence for Clifford algebras to the full setting of superrings, without requiring commutativity of the even part or the existence of a bilinear form on the odd part.

In this sense, our result unifies and generalizes the classical theory: every super-ring gives rise to a "Clifford-type" matrix ring, and the category of supermodules over the superring is always equivalent to the category of ordinary modules over this matrix ring. This provides a robust categorical framework that encompasses the classical examples while also covering new, noncommutative, and form-free situations.

7. TRIVIAL EXTENSIONS AS TRIANGULAR MATRIX RINGS

We now show that every trivial extension admits a natural representation as a ring of upper triangular matrices. This perspective not only clarifies the algebraic structure of the extension but also connects it naturally with the theory of superrings and, in special cases, with Clifford algebras.

Definition 7.1. *Let R be an associative ring with unity and let M be an R - R -bimodule. The **triangular matrix ring** associated to the pair (R, M) is the set*

$$T(R, M) := \left\{ \begin{pmatrix} r & m \\ 0 & s \end{pmatrix} \mid r, s \in R, m \in M \right\},$$

with addition defined componentwise and multiplication given by the usual matrix product:

$$\begin{pmatrix} r & m \\ 0 & s \end{pmatrix} \begin{pmatrix} r' & m' \\ 0 & s' \end{pmatrix} = \begin{pmatrix} rr' & rm' + ms' \\ 0 & ss' \end{pmatrix}.$$

It is straightforward to verify that $T(R, M)$ is an associative ring with identity $\begin{pmatrix} 1_R & 0 \\ 0 & 1_R \end{pmatrix}$.

The following result establishes the fundamental isomorphism.

Proposition 7.1. *Let R be a ring and M an R - R -bimodule. Then the trivial extension $R \ltimes M$ is isomorphic to the triangular matrix ring $T(R, M)$. Specifically, the map*

$$\Phi : R \ltimes M \longrightarrow T(R, M), \quad (r, m) \longmapsto \begin{pmatrix} r & m \\ 0 & r \end{pmatrix}$$

is a ring isomorphism.

Proof. The map Φ is clearly bijective, with inverse $\begin{pmatrix} r & m \\ 0 & s \end{pmatrix} \mapsto (r, m)$ (note that the entry s is forced to equal r in the image, so the inverse is well-defined). To see that Φ is a homomorphism, compute:

$$\Phi((r, m))\Phi((s, n)) = \begin{pmatrix} r & m \\ 0 & r \end{pmatrix} \begin{pmatrix} s & n \\ 0 & s \end{pmatrix} = \begin{pmatrix} rs & rn + ms \\ 0 & rs \end{pmatrix} = \Phi(rs, rn + ms) = \Phi((r, m)(s, n)).$$

Thus Φ preserves multiplication. It clearly preserves addition and the unit, so it is an isomorphism of rings. $\square$

The isomorphism above is sometimes written with the matrix $\begin{pmatrix} r & m \\ 0 & 0 \end{pmatrix}$ or with the entry $s = 0$, but the version with equal diagonal entries is the most common in the context of trivial extensions, as it reflects the fact that the extension is a ring with the same unit as R .

Now suppose that R is a superring and that M is an R - R -bimodule (not necessarily graded). We can endow the trivial extension with the superring structure described in Section ??, where

$$(R \ltimes M)_0 = R \oplus 0, \quad (R \ltimes M)_1 = 0 \oplus M.$$

Under the isomorphism Φ , this grading translates into a natural grading on the triangular matrix ring $T(R, M)$.

Proposition 7.2. *Let R be a superring and let M be an R - R -bimodule. The triangular matrix ring $T(R, M)$ is a superring with $\mathbb{Z}_2$ -grading given by*

$$T(R, M)_0 := \left\{ \begin{pmatrix} r & 0 \\ 0 & s \end{pmatrix} \mid r, s \in R \right\},$$

$$T(R, M)_1 := \left\{ \begin{pmatrix} 0 & m \\ 0 & 0 \end{pmatrix} \mid m \in M \right\}.$$

Moreover, the isomorphism $\Phi : R \ltimes M \rightarrow T(R, M)$ is an isomorphism of superrings.

Proof. We verify that the grading is compatible with the matrix multiplication. For two even matrices

$$A = \begin{pmatrix} r & 0 \\ 0 & s \end{pmatrix}, \quad B = \begin{pmatrix} r' & 0 \\ 0 & s' \end{pmatrix},$$

their product is

$$AB = \begin{pmatrix} rr' & 0 \\ 0 & ss' \end{pmatrix} \in T(R, M)_0.$$

For an even matrix A and an odd matrix $C = \begin{pmatrix} 0 & m \\ 0 & 0 \end{pmatrix}$, we have

$$AC = \begin{pmatrix} r & 0 \\ 0 & s \end{pmatrix} \begin{pmatrix} 0 & m \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & rm \\ 0 & 0 \end{pmatrix} \in T(R, M)_1,$$

and similarly,

$$CA = \begin{pmatrix} 0 & m \\ 0 & 0 \end{pmatrix} \begin{pmatrix} r & 0 \\ 0 & s \end{pmatrix} = \begin{pmatrix} 0 & ms \\ 0 & 0 \end{pmatrix} \in T(R, M)_1.$$

Finally, for two odd matrices C and D , their product is

$$CD = \begin{pmatrix} 0 & m \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & n \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \in T(R, M)_0.$$

Thus $T(R, M)_i T(R, M)_j \subseteq T(R, M)_{i+j}$ for all $i, j \in \mathbb{Z}_2$. The unit matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ lies in the even part. Therefore, $T(R, M)$ is a superring.

The isomorphism Φ sends $(r, 0)$ to $\begin{pmatrix} r & 0 \\ 0 & r \end{pmatrix} \in T(R, M)_0$ and $(0, m)$ to $\begin{pmatrix} 0 & m \\ 0 & 0 \end{pmatrix} \in T(R, M)_1$, so it preserves the grading and is an isomorphism of superrings. $\square$

This representation shows that the trivial extension is, in essence, a superring of upper triangular matrices whose diagonal entries lie in R and whose off-diagonal entries lie in M , with the odd part consisting precisely of the nilpotent matrices with zero diagonal. This is a very concrete and useful model for computations.

When R is a commutative ring and M is a projective R -module with a symmetric bilinear form, the triangular matrix ring $T(R, M)$ equipped with this grading is closely related to the Clifford algebra associated to the form. In fact, for certain

choices of the form, the Clifford algebra can be realized as a subring of a larger matrix ring, and the above construction recovers the even/odd decomposition of the Clifford superalgebra. This connection is explored in detail in [5] and [10].

The functoriality of the trivial extension extends naturally to this matrix representation. Indeed, if $f : R \rightarrow S$ is a superring homomorphism and $g : M \rightarrow N$ is a compatible bimodule homomorphism, then the induced map

$$T(R, M) \rightarrow T(S, N), \quad \begin{pmatrix} r & m \\ 0 & s \end{pmatrix} \mapsto \begin{pmatrix} f(r) & g(m) \\ 0 & f(s) \end{pmatrix}$$

is a morphism of superrings. Hence the assignment $(R, M) \mapsto T(R, M)$ is functorial, recovering the functor T_R defined earlier.

This triangular matrix model provides a powerful and intuitive way to view trivial extensions, and it will be particularly useful in the sequel when we study representations and homological properties of supermodules.

Throughout this work, we have encountered two distinct matrix representations associated to a superring $R = R_0 \oplus R_1$. On the one hand, we have the **triangular matrix ring**

$$T(R_0, R_1) := \begin{pmatrix} R_0 & R_1 \\ 0 & R_0 \end{pmatrix},$$

which represents the trivial extension $R_0 \ltimes R_1$ as a superring. On the other hand, we have the **parity matrix ring**

$$S := \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix},$$

which arose as the endomorphism ring of the canonical generator $G = R \oplus R^*$ of the category $s\mathcal{M}_R$ and, under suitable conditions, is interpreted as a Clifford algebra.

We now show how these two representations are intimately related and how the triangular matrix representation fits naturally into the broader framework of superrings and their Morita theory.

Proposition 7.3 (The Triangular Matrix Ring as a Superring). *Let $R = R_0 \oplus R_1$ be a superring, and let $M = R_1$ be regarded as an R_0 - R_0 -bimodule via the restrictions of the multiplication in R . Then the trivial extension $R_0 \ltimes R_1$ is isomorphic, as a superring, to the triangular matrix ring*

$$T(R_0, R_1) = \begin{pmatrix} R_0 & R_1 \\ 0 & R_0 \end{pmatrix},$$

with grading

$$T(R_0, R_1)_0 = \begin{pmatrix} R_0 & 0 \\ 0 & R_0 \end{pmatrix}, \quad T(R_0, R_1)_1 = \begin{pmatrix} 0 & R_1 \\ 0 & 0 \end{pmatrix}.$$

Proof. This is precisely Proposition 7.1 applied to the pair (R_0, R_1) , together with the grading described in the previous section. $\square$

In this representation, the odd part consists precisely of the strictly upper triangular matrices with zero diagonal. This reflects the fact that in the trivial extension, the product of two odd elements is zero: $(0, m)(0, n) = (0, 0)$. In matrix form, this corresponds to the product of two strictly upper triangular matrices being zero.

Now, recall that the full parity matrix ring

$$S = \begin{pmatrix} R_0 & R_1 \\ R_1 & R_0 \end{pmatrix}$$

is a superring with even part $\begin{pmatrix} R_0 & 0 \\ 0 & R_0 \end{pmatrix}$ and odd part $\begin{pmatrix} 0 & R_1 \\ R_1 & 0 \end{pmatrix}$. In this ring, the product of two odd matrices is generally nonzero:

$$\begin{pmatrix} 0 & m \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & n \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad (\text{in the triangular case}),$$

whereas in the parity matrix ring,

$$\begin{pmatrix} 0 & m \\ n & 0 \end{pmatrix} \begin{pmatrix} 0 & p \\ q & 0 \end{pmatrix} = \begin{pmatrix} mq & 0 \\ 0 & np \end{pmatrix}$$

which is generally nonzero. This captures the possibility of a nontrivial product between odd elements, which is absent in the trivial extension.

Proposition 7.4 (The Triangular Ring as a Sub-superring of the Parity Matrix Ring). *If the odd part R_1 is equipped with a product $R_1 \times R_1 \rightarrow R_0$ making R a superring (i.e., the original superring structure), then the triangular matrix ring $T(R_0, R_1)$ embeds naturally as a sub-superring of the parity matrix ring S via the inclusion*

$$\begin{pmatrix} r & m \\ 0 & s \end{pmatrix} \mapsto \begin{pmatrix} r & m \\ 0 & s \end{pmatrix},$$

where on the right-hand side we view the matrix as an element of S with the lower-left entry set to zero.

Proof. The inclusion is clearly injective and preserves addition and the grading. It preserves multiplication because the matrix product in S restricted to the upper triangular matrices coincides with the product in $T(R_0, R_1)$. However, note that the image is not closed under the full multiplication of S unless the odd part is trivial; it is only a subring with respect to the restricted multiplication. Thus, $T(R_0, R_1)$ is a sub-superring of S . $\square$

The significance of this connection is twofold. First, the triangular matrix representation provides a concrete and highly accessible model for the trivial extension, making computations transparent and explicitly showing the graded structure. Second, it reveals that the trivial extension is the "square-zero" case of the more general parity matrix ring: when the odd part has trivial multiplication, the full parity matrix ring collapses (in a suitable sense) to the triangular one. This is precisely the situation of the trivial extension.

Furthermore, this aligns with the interpretation of S as a Clifford algebra. As discussed in Section ??, a Clifford algebra is determined by a bilinear form on the odd part. When the bilinear form is zero, the Clifford algebra reduces to the exterior algebra, which in its matrix representation coincides with the triangular matrix ring. Thus, the triangular matrix representation is the matrix realization of the Clifford algebra associated to the zero quadratic form.

Corollary 7.1. *The triangular matrix ring $T(R_0, R_1)$ is a superring that serves as the universal square-zero extension of R_0 by R_1 . It is the minimal superring that contains R_0 as its even part and R_1 as its odd part, with the condition that the odd part squares to zero.*

From the categorical perspective, the triangular matrix representation is not merely a computational aid. It is functorial: any morphism of trivial extensions

induces a morphism of triangular matrix rings, and vice versa. Moreover, the Morita equivalence established earlier,

$$s\mathcal{M}_R \simeq \mathcal{M}_S,$$

specializes, in the case where $R_1^2 = 0$ and R_0 is the even part, to a Morita equivalence between the category of supermodules over the trivial extension and the category of ordinary modules over the triangular matrix ring $T(R_0, R_1)$. This connects all the threads of our discussion: the superring structure, the matrix representation, and the categorical equivalence.

For a comprehensive treatment of matrix representations of superrings and their applications, we refer to the standard texts on graded ring theory [13] and the works on Morita theory for superalgebras [7].

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FACULTAD DE CIENCIAS, UNAM, MEXICO CITY
Email address: `murphy@ciencias.unam.mx`

FACULTAD DE CIENCIAS, UNAM, MEXICO CITY
Email address: `@ciencias.unam.mx`